# Tharun V. Puthanveettil

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**Human-Inspired Robot Learning | Multimodal Learning | Human–Robot Interaction**

## Research Focus

I want to build robots that understand the people they work with and keep learning from them over time. I take inspiration from how people learn: we build on experience, recognize what a new situation requires, identify gaps in what we know, and acquire and retain useful knowledge. I want to investigate how robot-learning architectures can accumulate and use contextual knowledge across tasks, people, and environments, so that acquiring one skill makes the next easier to learn. My work spans multimodal human understanding, learning from intervention, policy explainability and predictive models, with physical-robot, simulation and offline evaluations.

## Research Threads

**Learning from People** PHYSICAL ROBOT AND SIMULATION

*How can robots improve from people's demonstrations and corrections without losing what they already do well?*

- Built a correction loop on a physical robot in which people demonstrate a task, take over when the policy fails, and have their corrections reviewed and fed back into training before the next policy is rolled out. Overview
- Gave people and AI models the same small, safety-checked actions for controlling the arm, so a human run and a policy run can be compared step by step and both reused for training.
- Developing a portable egocentric interface so demonstrations can come from people doing a task themselves, not only from teleoperation, through the same data pipeline. Overview
- Tested in simulation what a policy keeps once the privileged inputs that guided its training are taken away, comparing teacher-student training with learned auxiliary heads.

**Policy Explainability and Representation** SIMULATION AND PHYSICAL ROBOT

*What has a policy learned, and what shapes its actions?*

- Started with attention maps, Grad-CAM, input ablations and prompt comparisons to see what drives a policy, then moved to causal tests, since these maps show correlation rather than cause. Overview
- Found with activation patching and representation probes in simulation that a policy can learn which word refers to which object, yet its action head does not use that knowledge. Held across several training setups and three seeds.
- On a real bimanual robot, separated choosing the right object from completing the pick, and found that a policy with higher overall success can be no better at choosing the object it was asked for.
- Learned to rule out rival explanations before reporting a result. What looked like robustness to a tilted camera turned out to be a policy that was ignoring that camera.

**Predictive Models, Inverse Dynamics, and Transfer** OFFLINE, SIMULATION AND PHYSICAL ROBOT

*How can predicted outcomes and inverse dynamics support action learning on physical robots?*

- Trained image-based inverse dynamics in simulation to test whether actions inferred from video can replace recorded ones, comparing against recorded-label and executed-command controls. Also retrained a published video-action model's action head on teleoperation data and evaluated it offline. Overview
- Fine-tuned a world-action model in simulation and judged its predicted futures, instruction grounding and failure signals separately from task success, since a good prediction does not guarantee a good action.
- On a real bimanual robot, found that offline action error did not predict which policy would succeed on the task.

**Understanding People** PRIOR WORK | 2021-2024

*How can a robot recognize what people are doing and what they mean from incomplete, multimodal evidence?*

- Co-developed Track Anything Rapter, where a person picks a target by click, box, image or text and a drone follows it, using foundation-model perception and visual servoing.
- Built an exploratory policy that learns from a person's demonstration, speech and gesture together (NatSGD dataset, with Snehesh Shrestha, UMD).
- Showed in a graduate project that fusing multiple camera views makes pose-based action recognition and fall detection more robust to occlusion, raising F1 from 0.76 to 0.82. PoseFusion

## Education

**Master of Engineering in Robotics, GPA: 3.9/4.0** **Aug. 2022 – May 2024**
*University of Maryland, College Park* *Maryland, USA*

**Selected coursework:** Advanced Techniques in Visual Learning and Recognition; Perception for Autonomous Robots; Planning for Autonomous Robots; Control of Robotic Systems; Fundamentals for Artificial Intelligence and Deep Learning Framework.

**B.Tech. in Electronics and Communication Engineering, GPA: 8.59/10.0** **Jul. 2014 – Apr. 2018**
*Vellore Institute of Technology* *India*

## Research and Professional Experience

**Robotics Engineer – Robot Learning and Deformable Manipulation** **Jul. 2024 – Present**
*Optimal — assigned to General Motors R&D* *Michigan, USA*

- Architected and implemented a modular ROS 2 platform for bimanual robot learning, from teleoperated demonstration and synchronized data capture to training, simulation, staged testing, and physical deployment. Shared interfaces let devices, learning methods, and simulators change without rebuilding the workflow.
- Developed human-correction and step-level control workflows, enabling comparable human and policy rollouts and analysis of failures and interventions.
- Built foundation-model services for task-conditioned detection, segmentation, tracking, and pose servoing. Reusing embeddings and propagating masks over time cut perception latency by about 35% and sustained 30–40 FPS.
- Developed force-aware bimanual manipulation of deformable linear objects using impedance control and force-torque feedback, incorporating elastic-tension regulation to support consistent manipulation.
- Built a calibrated digital twin and a behavior-tree-driven synthetic-data workflow to test how policies degrade under changes in sensing, timing, contact, and appearance.
- Designed the Cartesian motion-control and safety layer for learned policies, and supervised its extension to compliant (admittance-controlled) dual-arm data collection and rollout.
- Compared action-chunking and open-source vision-language-action policies while developing evaluation measures beyond success rate (partial progress, failure type, recovery, repeatability).

**Researcher – AI and Robotics** **Feb. 2023 – May 2024**
*CATT Laboratory, University of Maryland* *Maryland, USA*

- Architected the fusion pipeline for multimodal anomaly detection, with separate LiDAR, odometry, and network encoders and reconstruction-based scoring. The published system achieved up to 98% accuracy with vector-based reconstruction loss versus 72% with scalar-based loss, and 97% with half the training data.
- Co-developed Track Anything Rapter, a ROS 2 aerial system for multimodal target specification (see Research Threads).

**Volunteer Researcher – Multimodal Human–Robot Interaction** **Oct. 2022 – Jun. 2023**
*Perception and Robotics Group (PRG), University of Maryland* *Maryland, USA*

- Implemented an exploratory demonstration, speech, and gesture policy (see Research Threads).
- Implemented a 3D human body mesh reconstruction pipeline using the SMPL model.

**Researcher – Collaborative Robotics and Computer Vision** **May 2021 – Dec. 2021**
*Indian Institute of Science* *Bangalore, India*

- Developed a vision-guided precision-weeding robot that estimates plant roots and spread, then plans with hybrid inverse kinematics to reach weeds while reducing contact with crops. First-author work, ICSR + AI 2025.
- Built synthetic-data and augmentation pipelines for weed localization, using hard-negative sampling and structured distractors to improve robustness under challenging backgrounds.
- Co-designed a dual-purpose end effector (plucking plus spraying or watering) and adaptive position/torque control for delicate plants.
- Developed and evaluated a U-Net monocular depth model in an IROS 2021 challenge setting.

**Project Engineer – AI Innovation** **Jul. 2018 – Apr. 2021**
*CTO Office, Wipro Digital* *Bangalore, India*

- Led applied-research prototypes in computer vision, HCI, recommendation, generative modeling, and mobile inference, and coordinated quarterly prototype selection and delivery.
- Built tool tracking, mobile vehicle recognition, photogrammetry, human-pose estimation, virtual try-on, and recommendation components for industrial and retail workflows.

### Selected Earlier Technical Projects

**Implicit Neural Representation and SLIC Segmentation** | *PyTorch, TensorFlow, OpenCV* 2023

- Implemented coordinate-conditioned image outpainting and deep-feature superpixel segmentation, including controlled comparisons with simpler baselines.

**Maze Runner and Autonomous Retrieval Robot** | *ROS 2, Nav2, visual servoing* 2023

- Built visual-cue-based maze navigation in Gazebo and an onboard-perception retrieval robot using sensor fusion and visual servoing.

**Reinforcement-Learning Cricket Simulation** | *adaptive agents and simulation* 2020

- Built a simulated bowler agent that adapts to batting styles and field configurations. code

## Publications

*(Also published as V. P. Tharun.)*

1. A. Zoulkarni, M. Noorani, J. S. Baras, J. Mirenzi, **T. V. Puthanveettil**, and C. D. Grody. "Efficient Integration of Domain Knowledge and Data-Driven Methods for Securing Autonomous Cyber-Physical Systems." IEEE ROSE 2026. DOI: 10.1109/ROSE69181.2026.11570974.
2. **V. P. Tharun** and A. R. Chowdhury. "Vision Based Hybrid IK Task Planning with Feedforward Neural Network for Collaborative Plant-Robot Interaction in Precision Farming." *International Conference on Social Robotics + AI (ICSR + AI 2025)*, LNAI 16132, pp. 253–267. Springer, 2026. DOI: 10.1007/978-981-95-2382-5_18.
3. M. Noorani, **T. V. Puthanveettil**, A. Zoulkarni, J. Mirenzi, C. D. Grody, and J. S. Baras. "Multimodal Anomaly Detection for Autonomous Cyber-Physical Systems Empowering Real-World Evaluation." *GameSec 2024*, LNCS 14908, pp. 306–325. DOI: 10.1007/978-3-031-74835-6_15.
4. L. P. Pratap, P. M. Shailendrasingh, A. Anand, and **V. P. Tharun**. "Wall-Climbing Robot Using Soft Robotics." *ICPCSI*, 2017.

Earlier work in data mining and forecasting (2018–2020): five papers and reports on rainfall forecasting and water-quality assessment. Google Scholar

## Preprints and Research Outputs

1. **T. V. Puthanveettil**, A. Singh, Y. Jain, V. Bukka, and S. Arjun S. "Auto-Platoon: Freight by Example." *arXiv:2405.11659*, 2024. code
2. **T. V. Puthanveettil** and F. Obaid ur Rahman. "Track Anything Rapter (TAR)." *arXiv:2405.11655*, 2024. code + demos
3. **T. V. Puthanveettil**, A. Singh, and A. Balram. "Pose Fusion - Multi-View Pose Integration for Comprehensive Action Recognition." UMD graduate course project. Project poster and preprint. preprint
4. **V. P. Tharun** et al. "Application of Mobile Collaborative Robot using Deep Learning in Precision Weed Control." Manuscript, 2021.
5. **V. P. Tharun** et al. "Event-Based Dynamic Obstacle Avoidance in Outdoor Environments." IROS 2021 workshop poster.

## Technical Skills

**Robot learning:** PyTorch, TensorFlow, LeRobot, action-chunking policies, vision-language-action models, diffusion/flow policies, imitation learning, self-supervised learning

**Robotics and control:** ROS 2, ROS, MoveIt 2, impedance/compliance control, visual servoing, behavior trees, bimanual manipulation, force-torque feedback

**Simulation and perception:** Isaac Sim/Lab, MuJoCo, Gazebo, OpenCV, Open3D, RGB-D cameras, detection, segmentation, tracking, pose estimation

**Programming and systems:** Python, C++, SQL, Linux, Docker, Git, REST APIs

## Selected Recognition

**3rd place, GNSS-denied navigation benchmark** — RAMI Cascade Campaign for Aerial Robots, IROS 2021, as part of team RIL-IISc. **Results**

**NRF 2020 showcase:** Led a three-person team developing a retail solution showcased at the National Retail Federation's Retail's Big Show.

**Top-two proposal selection, 2019:** Proposed a solution for a Japanese company's digital elder-care initiative.

## Intellectual Property, Mentoring, and Communication

**Patents:** Inventor on 11 invention disclosures at General Motors, two on the patent route and three protected as trade secrets; details confidential.

**Research supervision:** Supervised an engineering intern extending the robot-learning platform for compliant bimanual data collection, learning, analysis, and rollout.

**FIRST Robotics:** Electrical Programming Head Coach, The Irrational Engineers (Jun.–Jul. 2024; occasional help since); mentored high-school students in controls, computer vision, and software.

**Writing:** *Human in the Loom*, essays on robot learning, embodied intelligence, and credible research practice. Essay archive